

Package ‘CAnetwork’

April 30, 2026

Title Ecological Networks Analysis with Correspondence Analysis

Version 1.2.0

Description Analyze trait matching and interaction niches in ecological networks using correspondence analyses.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Imports ade4,
adegraphics,
dplyr,
ggplot2,
ggrepel,
grid,
mvtnorm,
patchwork,
rphylopic,
stats,
stringr,
tibble,
tidyr,
tidyselect

Suggests svglite

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compas2d

*Simulate a community matrix***Description**

Simulate abundance values based on species traits. The ecological interpretation can be a sites x species matrix or a resource x consumer species interaction matrix.

Usage

```
compas2d(
  nsp = 40,
  le.grad = 100,
  ratio.grad = 0.8,
  nsite = 100,
  col_abund = NULL,
  row_abund = NULL,
  col_prefix = "s",
  row_prefix = "e",
  rowvar_prefix = "env",
  remove_zeroes = FALSE,
  buffer = 1,
  min.ab = 1,
  max.ab = 100,
  ninter = 100,
  delta = 1,
  mean.tol = 2,
  sd.tol = 10,
  return_intermediate = FALSE
)
```

Arguments

nsp	Number of species / consumer
le.grad	gradient length for traits/environmental variables. The values are comprised between 0 and le.grad.
ratio.grad	length of the second gradient as a ratio of the first one.
nsite	Number of sites / resource species
col_abund	Optional vector of abundances for species/consumers. If missing, abundances are taken from a uniform law between min.ab and max.ab.
row_abund	Optional vector of abundances in sites/ for resource species. If missing, abundances are taken from a uniform law between min.ab and max.ab.
col_prefix	Prefix for the column names of the matrix. Defaults to "s" for "species".
row_prefix	Prefix for the row names of the matrix. Defaults to "e" for "environment".
rowvar_prefix	Prefix for the column names of the row variable. Defaults to "env" for "environment".
remove_zeroes	If there are unobserved species, should we keep them in the final matrix?

buffer	buffer to allow species / consumer species optima to fall outside the gradient by 0 - buffer ; le.grad + buffer.
min.ab	minimal abundance value for the uniform law (if col_abund and/or row_abund are absent)
max.ab	maximal abundance value for the uniform law (if col_abund and/or row_abund are absent)
ninter	Total number of observations in the web
delta	Exponent between 0 and 1 to give more (1) or less (0) weight to trait matching relatively to abundance.
mean.tol	Mean value for the species/consumers niche tolerance (standard deviation). The species/consumers niche tolerance are drawn from a normal law of mean mean.tol and standard deviation sd.tol for each trait axis.
sd.tol	Standard deviation for the species/consumers niche tolerance (standard deviation) The species/consumers niche tolerance are drawn from a normal law of mean mean.tol and standard deviation sd.tol for each trait axis.
return_intermediate	Return intermediate results? If yes, will return the probability of interaction based on matching only, on abundances only and the predicted mix of matching and abundances (before sampling).

Value

An object of class `compas`, which is a list with 3 components (4 if `return_intermediate`):

- `comm`: the community matrix of dimension `nsite` x `nsp`. Row names are `row_prefix` followed by `1..nsite` and column names are `col_prefix` followed by `1..nsp`.
- `rowvar`: the values of the row variables (resource traits/environment gradient values). It is a `nsite` x 2 matrix. Row names are `row_prefix` followed by `1..nsite` and column names are `rowvar_prefix` followed by 1 and 2.
- `colvar`: the values of the column variables (consumer/species traits values). It is a `nsp` x 2 x 2 array. Row names are `row_prefix` followed by `1..nsp`. Column names in each array dimension are `mean` and `sd` (for the mean and standard deviation of the species niche, respectively). Each array 3rd dimension corresponds to a resource trait or environmental gradient value.
- if `return_intermediate`, a fourth component named `intermediate` is returned. The first element is named `p_matching` and contains the matrix of probabilities of interactions based only on matching. The second `ab_neutral` contains the matrix of count of interactions as predicted by species abundances. The third `p_mix` contains the matrix of probabilities of interactions taking into account the mix of matching and abundances.

create_code

Create code

Description

Create a code for each (unique) value of a vector

Usage

```
create_code(col, code_pattern = NA, name = NA)
```

Arguments

col	column to code
code_pattern	first letter to add before code
name	optional name for output df

Value

dataframe with 2 columns: one with original vector, the other with the corresponding codes

df_to_matrix	<i>Dataframe to matrix</i>
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Description

Transforms a dataframe into an incidence matrix.

Usage

```
df_to_matrix(df, rows = "plant_species_code", columns = "animal_species_code")
```

Arguments

df	the dataframe to transform. Must have column names indicated in rows and columns and a third column frequency with the interaction frequency/strength index.
rows	Name of the column with the species names that will be the rows of the matrix.
columns	Name of the column with the species names that will be the columns of the matrix.

Value

Returns an incidence matrix with n rows and m columns filled with the values in frequency.

filter_matrix	<i>Filter matrix</i>
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Description

Filter incidence matrix to keep only species that interact with more partners than a given threshold.

Usage

```
filter_matrix(mat, thr)
```

Arguments

mat	The incidence matrix.
thr	The minimal number of different interacting partners a species must have (a species must have $\geq$ thr different interacting partners). It is not weighted (for instance, if thr = 2 then a species interacting 100 times with a unique other partner will be removed.)

Value

The interaction matrix but where the rows and/or columns with species that do not interact above the threshold removed.

multiplot	<i>Plot bi, tri or quadriplots</i>
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Description

Bi, tri or quadriplots from a multivariate analysis

Usage

```
multiplot(
  indiv_row = NULL,
  indiv_col = NULL,
  indiv_row_lab = rownames(indiv_row),
  indiv_col_lab = rownames(indiv_col),
  var_row = NULL,
  var_col = NULL,
  var_row_lab = rownames(var_row),
  var_col_lab = rownames(var_col),
  row_color = "black",
  col_color = "black",
  eig = NULL,
  x = 1,
  y = 2,
  xlim = NULL,
  ylim = NULL,
  grad = 4,
  mult = 1,
  title = NULL,
  max.overlaps = 20,
  force_pull = 1,
  force = 1,
  labsize = 3,
  alphapoints = 0.8
)
```

Arguments

indiv_row	Matrix of individuals coordinates for rows (type dudi\$li)
indiv_col	Matrix of individuals coordinates for columns (type dudi\$co)
indiv_row_lab	Labels for row individuals
indiv_col_lab	Labels for columns individuals
var_row	Matrix of variables coordinates for rows (type dudi\$corR)
var_col	Matrix of variables coordinates for columns (type dudi\$corQ)
var_row_lab	Labels for row variables
var_col_lab	Labels for columns variables
row_color	Color for the row individuals or variables
col_color	Color for the columns individuals or variables
eig	Eigenvalues vector
x	Axis to use for the x-axis
y	Axis to use for the y-axis
xlim	x-axis limits. Defaults to data range if not specified
ylim	y-axis limits. Defaults to data range if not specified
grad	graduations for the major.grid
mult	factor to multiply the vectors (arrows on the plot)
title	plot title
max.overlaps	max.overlaps argument for ggrepel::geom_text_repel
force_pull	force_pull argument for ggrepel::geom_text_repel
force	force argument for ggrepel::geom_text_repel
labsize	text size for point labels
alphapoints	Transparency value for points

Value

a ggplot

patch_plot

Patch plot

Description

Arrange plots with patchwork

Usage

```
patch_plot(
  p_data,
  p_resource,
  p_consumer,
  n_consumer,
  n_resource,
  traits_dim = 1
)
```

Arguments

p_data	data matrix plot (generated with plot_data())
p_resource	resource traits plot (generated with plot_traits())
p_consumer	consumer traits plot (generated with plot_traits())
n_consumer	number of resources for plot height
n_resource	number of consumers for plot width
traits_dim	width/height of the traits matrix (use 1 for ordination, 2 for data)

Value

A patchwork ggplot

plotmat	<i>Plot matrices for the model</i>
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Description

Function to plot matrices Y, E (environment/resource traits) and T (species traits/consumer traits)/
This function plots the matrices arranged in a square, with the user-given proportions.

Usage

```
plotmat(
  mar = 2,
  marl = 1.2,
  E = FALSE,
  T_ = FALSE,
  r,
  c,
  l = NULL,
  k = NULL,
  name_mat = TRUE,
  name_dim = TRUE,
  Yname = "italic(Y)",
  Ename = "italic(E)",
  Tname = "italic(T)",
  rname = "italic(r)",
  cname = "italic(c)",
  horizr = TRUE,
  plot.margin.x = 1,
  plot.margin.y = 1
)
```

Arguments

mar	horizontal margin between matrices
marl	vertical margin between matrices
E	Plot matrix E (rows variables)?

T_	Plot matrix T (columns variables)?
r	number of rows of matrix Y
c	number of columns of matrix Y
l	number of columns of matrix E
k	number of columns of matrix T
name_mat	display matrices names?
name_dim	display dimension names?
Yname	name of the matrix Y (center matrix)
Ename	name of the matrix E (row variables matrix)
Tname	name of the matrix T (column variables matrix)
rname	name of the dimension r
cname	name of the dimension c
horizr	display r dimension horizontally? (else, will be vertical)
plot.margin.x	horizontal margin around plot
plot.margin.y	vertical margin around plot

Value

a ggplot of matrix Y (and optionally E and T), with their dimensions written above and aligned in a square where the fourth-corner is missing.

Examples

```
plotmat(r = 5, c = 2)
```

plot_corcircle	<i>Plot correlation circle</i>
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Description

Plot the correlation circle

Usage

```
plot_corcircle(
  cor,
  cor2 = NULL,
  xax = 1,
  yax = 2,
  xlab = NULL,
  ylab = NULL,
  label = rownames(cor),
  eig = NULL,
  mar = 0.01,
  labsize = 3.88,
  col_bg = "grey30",
  col = "black",
```



```

    lty = "solid",
    col2 = "black",
    lty2 = "solid",
    label2 = rownames(cor2),
    lim = 1,
    clip = "off"
  )

```

Arguments

cor	dataframe containing the coordinates of the variables for each axis in columns
cor2	optional second set of variables (for double constrained analyses)
xax	index of the column of the matrix to use for x
yax	index of the column of the matrix to use for y
xlab	Custom x-label
ylab	Custom y-label
label1	label to display on the arrows
eig	Eigenvalues vector
mar	margin between arrow tips and label
labsize	Size for arrow labels.
col_bg	color to use for background elements (circle and $y = 0$ and $x = 0$)
col	color for arrows (unique)
lty	linetype for the first set of variables
col2	optional color for the second set of variables
lty2	linetype for the second set of variables
label2	optional label to display on the arrows for the second set of variables
lim	limits for x and y: a unique number that gives the distance between the plot limit and zero (so that the circle is centered)
clip	constrain the labels to be inside the plot? If yes, change to "on".

Value

A ggplot with the correlation circle where variables are represented as arrows inside the circle of radius 1.

plot_eig

Plot eigenvalues

Description

Barplot of eigenvalues

Usage

```
plot_eig(eigenvalues, showrank = FALSE)
```

Arguments

eigenvalues	The eigenvalues
showrank	Should the x-axis display the rank of the eigenvalues?

Value

A ggplot

plot_table	<i>Plot table</i>
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Description

Plot a matrix

Usage

```
plot_table(
  x,
  y,
  abundance_data,
  abund_table,
  label_consumer = NULL,
  img_consumer = NULL,
  WA = FALSE
)
```

Arguments

x	consumers abundance vector
y	resources abundance vector
abundance_data	abundance data table: expected to have columns
abund_table	table corresponding to abundance_data (to wide format = interaction matrix) Consumer_sp, Resource_sp, Abundance and coa_c and coa_r of WA is TRUE.
label_consumer	label of consumer species (only useful if WA is TRUE)
img_consumer	image of consumer species (only useful if WA is TRUE)
WA	display the weighted averaging for one species?

Value

a ggplot

plot_traits	<i>Plot traits</i>
-------------	--------------------

Description

Plot trait matrices

Usage

```
plot_traits(traits, order, label, img, fill = "grey", type = "consumer")
```

Arguments

traits	traits dataframe (expected to have a size column)
order	order to use for the species
label	species labels
img	species image
fill	fill color (use a unique color to ignore the size)
type	either "resource" (row) or "consumer" (column)

Value

a ggplot

plot_traits_data	<i>Plot traits data</i>
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Description

Plot trait data matrices

Usage

```
plot_traits_data(traits, label, img, fill, type = "consumer", reorder = FALSE)
```

Arguments

traits	traits dataframe (expected to have a size column)
label	species labels
img	species image
fill	fill color (use a unique color to ignore the size)
type	either "resource" (row) or "consumer" (column)
reorder	should species be reordered by label?

Value

a ggplot

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